

Load-Dependent Sacroiliac Joint Micromotion in a Large Pelvic Cohort: Directional Re-routing, Between-Subject Variability and Scaling

Petr Henyš^{1,*}, Niels Hammer²

¹Institute of New Technologies and Applied Informatics,
Faculty of Mechatronics, Informatics and Interdisciplinary Studies,
Technical University of Liberec, Studentská 1402/2, 461 17 Liberec, Czech Republic

²Division of Macroscopic and Clinical Anatomy,
Gottfried Schatz Research Center, Medical University of Graz,
Auenbruggerplatz 25, 8036 Graz, Austria

*Corresponding author: Petr Henyš (petr.henys@tul.cz)

Abstract

Background: Sacroiliac joint (SIJ) micromotion depends on pelvic anatomy, material properties, and load direction, yet large-cohort evidence under standardized standing and parturition-motivated loading remains limited, particularly for separating geometric versus material sources of between-subject variability.

Methods: We analyzed a precomputed finite-element repository of 278 pelves (150 female, 128 male; age 16–91 years) represented on a common template via an implicit subject-specific mapping. Five standardized static load cases were studied: two-leg standing (SP2leg), one-leg standing (SP1leg), and three parturition-motivated proxies (LAB1–LAB3). Outcomes were SIJ rotation/translation magnitudes and AP/ML/CC translation components. We used robust and cluster-robust regression with adjustment for age, total pelvic volume, and three a priori pelvic dimensions (AP diameter, biischadic width, subpubic angle), plus variance-channel ratios (shape-only/full and material-only/full) and log–log size-scaling models.

Results: SP1leg produced the largest SIJ motion (median rotation 2.52°; translation 1.76 mm). Relative to SP2leg, LAB loads reduced translation magnitude but showed clear directional re-routing of AP/ML/CC components. After coarse anatomical adjustment, female sex was associated with higher LAB translation magnitude (0.20–0.23 mm across

LAB1–LAB3) and higher LAB3 rotation. Between-subject variability was overwhelmingly geometry-dominant (shape/full 98.7–102.1%), whereas material-only variance fractions were small (0.001–0.317%). Negative size scaling was clearest for SP1leg and for pooled translation models; sex-by-size slope interactions were not robust after false-discovery correction.

Conclusions: Under standardized forces, SIJ response is strongly load dependent, and between-subject dispersion is dominated by pelvic geometry rather than material mapping. Parturition-motivated loads alter the directional partitioning of SIJ motion more clearly than they increase its scalar magnitude. Size scaling is load dependent, with the clearest negative associations under one-leg standing and in pooled translation models.

Keywords: sacroiliac joint; pelvic anatomy; finite-element analysis; sexual dimorphism; scaling; parturition biomechanics

1 Introduction

The sacroiliac joint (SIJ) is a mechanically demanding articulation of the pelvic ring: it must transmit loads between the trunk and lower limbs while permitting only small, tightly constrained motions (Vleeming et al., 2012; Snijders et al., 1993). That combination—effective load transfer with low mobility—is enabled by irregular articular surfaces, strong ligamentous restraint, and the closed-ring organization of the pelvis (Vleeming et al., 2012; Snijders et al., 1993). More broadly, the human pelvis sits at the intersection of locomotor and obstetric demands, a framing that continues to motivate discussions of pelvic form and function in anatomy, anthropology, and biomechanics (Washburn, 1960; Whitcome et al., 2017; Gruss and Schmitt, 2015; Ruff, 2017).

Because these motions are small, they are difficult to measure consistently in vivo and across experimental setups. Reported SIJ kinematics vary with both task and measurement method, and the literature emphasizes that the joint should be treated as a low-motion but load-sensitive structure rather than a rigid articulation (Goode et al., 2008; Meijer et al., 2014; Keizer et al., 2019; No"e et al., 2020; Vleeming et al., 2012). Crucially, SIJ motion should not be expected to scale uniformly across directions: translation components in the anteroposterior (AP), medial-lateral (ML), and craniocaudal (CC) axes can change disproportionately with loading direction and constraint, reflecting anisotropic joint anatomy and the ligamentous network of the pelvic ring (Vleeming et al., 2012; Goode et al., 2008; Meijer et al., 2014; No"e et al., 2020).

Directional behavior becomes most informative when loading changes. Unilateral support

in locomotion increases asymmetry compared with quiet two-leg standing, and instrumented studies show that routine activities can generate large, task-dependent hip forces that must be routed through the pelvic ring (Bergmann et al., 2001; Kutzner et al., 2016). In obstetric contexts, pelvic anatomy and loading are additionally shaped by parturition-related functional demands, including uterine contraction and intra-abdominal pressure environments that differ across pregnancy and labor (House et al., 2022; Fuchs et al., 2013; Chun and Kirkpatrick, 2012; Mora et al., 2022). Pelvic form varies substantially between individuals and between sexes, including dimensions that relate to inlet, midplane, and outlet architecture (Ruff, 2017; Betti and Manica, 2020; Weber et al., 2019; Whitcome et al., 2017; Weiss, 2020). Such variation can shift lever arms, articular orientation, and ligament attachment geometry, and therefore may alter how otherwise standardized external forces are redistributed into SIJ micromotion. Beyond sex, skeletal tissues remodel across the lifespan, suggesting that age-related changes in geometry and material behavior may further modulate joint-level response (Farr and Khosla, 2015; Zebaze et al., 2010).

Finite-element (FE) modelling provides a complementary route to SIJ biomechanics because it can combine subject-specific anatomy with controlled boundary conditions and precisely defined load cases. Prior SIJ and pelvic-ring FE studies show substantial sensitivity to ligament restraint, cartilage properties, and loading configuration (Anderson et al., 2020; Yang et al., 2020; Song et al., 2021; Calixte et al., 2022). Computational childbirth models likewise highlight both the promise and the complexity of obstetric pelvic mechanics, including the interplay of geometry, tissue properties, and load direction, but many models are single-subject or focus on contact-rich scenarios rather than population-scale joint micromotion (Chen and Grimm, 2021; Roque de Almeida et al., 2024; Mora et al., 2022). More generally, SIJ FE studies are often limited by cohort size, by geometric normalization that removes absolute size information, or by uncertainty about whether between-subject dispersion is driven primarily by geometry (shape) versus material mapping.

Increasingly, large-cohort FE workflows adopt a common reference domain and encode subject-specific anatomy through an implicit mapping rather than explicit per-subject remeshing. This representation improves computational scalability, but it also shifts how “geometry” is represented and how size-related covariates should be interpreted: individual shape is carried by the mapping, even when the computational mesh is shared.

Two gaps are especially important for interpreting cohort-scale SIJ micromotion. First,

variance in predicted motion can arise from at least two mechanistic “channels”: geometric variability (including joint orientation and pelvic ring architecture) and material variability (e.g., density-based bone stiffness mapping) (Carter and Hayes, 1977; Keller, 1994; Morgan et al., 2003; Frost, 1987; Huiskes et al., 1987). Distinguishing these channels is necessary to determine whether anatomy alone largely governs *between-subject variability* under standardized loads, or whether plausible material heterogeneity contributes comparably. Second, size scaling of SIJ micromotion remains underreported in large cohorts. Under force-standardized loading, larger load-bearing geometry may reduce micromotion, and global size can confound sex comparisons given well-established sex-related pelvic shape differences (Ruff, 2017; Betti and Manica, 2020; Weber et al., 2019; Weiss, 2020). In reference-domain workflows, this requires a consistent definition of “size” derived from the same implicit geometry representation as the FE solve.

To address these issues, we analyzed a large precomputed FE repository of 278 pelvises in which subject geometry is represented on a common template through an implicit displacement mapping ϕ . The repository includes full, shape-only, and material-only variants, enabling variance-channel decomposition under identical standardized load definitions. We evaluated SIJ rotations and translations under five static load cases spanning locomotor and parturition-motivated proxies (two-leg standing, one-leg standing, and three labor-motivated loading states). By combining directional (AP/ML/CC) outcomes with robust and cluster-robust regression, we aimed to quantify load-dependent directional re-routing, sex effects under parturition-motivated loading after coarse anatomical adjustment, the relative contribution of shape versus material variance channels, and power-law size scaling of SIJ micromotion using total pelvic volume as a global proxy. We therefore tested the following pre-specified hypotheses.

Hypotheses

We pre-specified four mechanistic hypotheses, each phrased as an explicit, testable prediction:

1. **H1 (load dependence and re-routing):** under identical standardized resultant forces, SIJ micromotion exhibits load-case-specific patterns. Compared with bilateral standing, unilateral standing increases overall motion magnitude, whereas parturition-motivated loading produces non-uniform changes across AP/ML/CC translation components (directional re-routing) rather than uniform scaling of all components.
2. **H2 (sex effects conditional on anatomy):** under parturition-motivated loads, female

sex is associated with greater SIJ micromotion after adjustment for age, a global size proxy, and a parsimonious set of pelvic dimensions chosen to capture inlet depth, outlet breadth, and pubic-arch opening.

3. **H3 (geometry-dominant variability):** between-subject variability in SIJ outcomes is dominated by geometric differences: shape-only simulations recover most of the full-model variance, whereas material-only perturbations explain a comparatively small fraction under the standardized loading protocol.

4. **H4 (size scaling):** SIJ micromotion follows load-dependent power-law scaling with global pelvic size. Under force-standardized standing loads, negative scaling is expected to be most evident in higher-demand support conditions, and potential sex differences in scaling are evaluated via an interaction term rather than assumed a priori.

2 Materials and methods

2.1 Repository and cohort

We used precomputed outputs from a finite-element pelvic repository and two matched auxiliary variants designed to isolate shape-only and material-only effects. The cohort included 278 subjects (150 female, 128 male). Demographic metadata (age, sex), pelvic morphology (eight landmark-based dimensions), SIJ kinematics, and geometry-derived global metrics (volume, surface area, scale factor) were provided as structured tables accompanying the repository. All analyses were scripted and outputs were generated directly from these tables without manual value editing.

The morphology table contained eight pelvic dimensions: AP diameter, biacetabular width, biischadic width, bituberous width, iliopectineal eminence width, pelvic inlet transverse (PIT), sacral width, and subpubic angle. Missingness for demographic, morphometric, global size-proxy, and SIJ primary outcomes was zero.

Ethical considerations

This study analyzed a previously published, anonymized imaging-derived dataset and its associated precomputed finite-element repository (BoneDat; (Henyš and Kuchař, 2025)). No new human imaging or interventions were performed for this work. According to the BoneDat de-

scriptor, use of the source imaging data was approved by the Ethics Committee of the University Hospital Hradec Králové (reference 202411IO3P) with waiver of informed consent; we refer readers to the BoneDat publication for full ethical and governance details.

Table 1: Cohort characteristics and pelvic dimensions. Values are mean $\pm$ SD unless noted.

Variable	Overall	Female	Male	Missing
N	278	150	128	0
Age (years)	53.90 $\pm$ 19.33	53.36 $\pm$ 19.74	54.52 $\pm$ 18.91	0
Inlet AP diameter (mm)	117.04 $\pm$ 10.52	121.04 $\pm$ 10.02	112.34 $\pm$ 9.07	0
Biacetabular width (mm)	125.81 $\pm$ 8.53	129.08 $\pm$ 8.21	121.98 $\pm$ 7.22	0
Biischiadic width (mm)	104.27 $\pm$ 9.33	109.87 $\pm$ 7.13	97.71 $\pm$ 7.06	0
Bituberous width (mm)	132.24 $\pm$ 9.15	136.46 $\pm$ 8.29	127.30 $\pm$ 7.49	0
Iliopectineal eminence width (mm)	102.38 $\pm$ 7.75	106.20 $\pm$ 7.02	97.90 $\pm$ 5.96	0
Pelvic inlet transverse (mm)	131.58 $\pm$ 8.88	134.85 $\pm$ 8.59	127.75 $\pm$ 7.61	0
Sacral width (mm)	117.08 $\pm$ 6.08	116.75 $\pm$ 5.67	117.47 $\pm$ 6.54	0
Subpubic angle (deg)	78.12 $\pm$ 10.12	85.80 $\pm$ 6.14	69.11 $\pm$ 5.26	0
Global scale factor (-)	0.96 $\pm$ 0.06	0.92 $\pm$ 0.05	1.00 $\pm$ 0.05	0
Total pelvic volume (cm ³)	823.58 $\pm$ 149.32	736.79 $\pm$ 105.99	925.28 $\pm$ 126.99	0
Total pelvic surface (cm ²)	1568.63 $\pm$ 169.26	1479.09 $\pm$ 133.64	1673.57 $\pm$ 144.80	0

2.2 Geometry, material mapping, and boundary conditions

The FE workflow uses a small-strain isotropic elastic formulation posed on a common reference mesh (Bonet and Wood, 1997; Marsden and Hughes, 1983). Subject-specific anatomy is introduced implicitly through a displacement mapping $\phi(\mathbf{X})$ that maps reference coordinates $\mathbf{X}$ to subject coordinates $\mathbf{x} = \mathbf{X} + \phi(\mathbf{X})$. This mapping is used in the elastic operators and to pull back loads defined on subject surfaces to the reference domain (Supporting Information: Supplementary Methods). Global size proxies (total volume, surface area, and a derived scale factor) were computed from the mapped geometry and provided alongside the kinematic outputs.

Material assignment followed repository defaults: bone modulus from density via a piecewise density–modulus relationship grounded in the broader bone mechanics literature (Carter and Hayes, 1977; Keller, 1994; Morgan et al., 2003; Frost, 1987; Huiskes et al., 1987),

$$E = \begin{cases} 2398 \text{ MPa}, & \rho_{ash} \leq 0.486 \\ 10200 \rho_{ash}^2, & \rho_{ash} > 0.486 \end{cases}, \quad \rho_{ash} = \frac{\rho + 0.09}{1.14}, \quad (1)$$

with bone Poisson ratio 0.3. SIJ cartilage and pubic symphysis regions used modulus 1.0 MPa and Poisson ratio 0.45, consistent with prior SIJ FE parameterizations emphasizing sensitivity to cartilage and ligament assumptions (Anderson et al., 2020; Yang et al., 2020; Calixte et al.,

2022). Shape and density fields were mapped with radial basis interpolation (smoothing 50, 10 neighbors). Boundary restraint was imposed on the S1 facet using penalty enforcement (surface penalty $\gamma = 10^6$).

2.3 Load cases

Five standardized static load cases were analyzed: SP2leg, SP1leg, LAB1, LAB2, LAB3 (Table 2). Forces were applied as total-force vectors on named anatomical surfaces, scaled by loaded-surface area in each subject so that resultant force vectors were identical across subjects. Standing loads were motivated by instrumented hip-force literature, whereas the LAB configurations were treated as simplified obstetric-mechanical proxies rather than full reconstructions of labor physiology (Bergmann et al., 2001; Kutzner et al., 2016; House et al., 2022; Mora et al., 2022).

Table 2: Standardized load-case definitions used for SIJ analysis.

Load case	Mechanical proxy	Applied total forces	Resultant
SP2leg	Bilateral standing	[0,0,+400] N on right AC notch and [0,0,+400] N on left AC notch	800 N cranio-caudal
SP1leg	Unilateral standing	[0,0,+800] N on right AC notch	800 N cranio-caudal
LAB1	Mediolateral ring compression proxy	[+400,0,0] N on ring_contact_left and [-400,0,0] N on ring_contact_right	0 N net; bilateral ML pair
LAB2	Ischial loading / SIJ distraction proxy	[+400,0,0] N on left_ischium_tuber and [-400,0,0] N on right_ischium_tuber	0 N net; bilateral ML pair
LAB3	Outlet AP distraction proxy	[0,-400,0] N on pubis_ins and [0,+400,0] N on SCJ	0 N net; AP pair

2.4 SIJ kinematic outcomes

For each load, SIJ outputs were available for left and right joints as three rotations (nutation, AP-axis rotation, CC-axis rotation; degrees) and three translations (ML, AP, CC; mm). Primary outcomes were:

1. mean left-right 3D rotation magnitude (deg),
2. mean left-right 3D translation magnitude (mm).

Secondary outcomes were absolute AP/ML/CC translation components. Additional derived outputs were rotational components and left-right asymmetry magnitudes. Mathematical and algorithmic details of the implicit geometry mapping, load pullback, SIJ rigid-body extraction from deformed geometry, derived scalar outcomes, and size-scaling exponent estimation are provided in the Supporting Information (Supplementary Methods).

2.5 Statistical models

Load-profile and directional re-routing (H1)

We summarized outcomes by load using medians and interquartile ranges. For directional re-routing, we computed within-subject differences relative to SP2leg for AP/ML/CC translation components and tested median shifts with Wilcoxon signed-rank tests (Wilcoxon, 1945); 95% CIs were bootstrap percentile intervals (Efron and Tibshirani, 1994).

Sex effects under LAB loads (H2)

For each LAB phase and each primary outcome, we fit robust linear models (HC3) (White, 1980; MacKinnon and White, 1985):

$$Y \sim \text{sex} + \text{age}_z + \log(\text{volume})_z + \text{AP}_z + \text{Biischiadic}_z + \text{Subpubic}_z + \text{sex} \times \log(\text{volume})_z. \quad (2)$$

The three pelvic dimensions were selected *a priori* as a parsimonious adjustment set representing inlet depth (AP), posterior/outlet transverse breadth (biischiadic width), and pubic-arch opening (subpubic angle). We preferred this compact set to an eight-dimension model because several pelvic widths were strongly correlated and a fully saturated morphometric adjustment would have reduced interpretability without constituting full-shape control. The sex coefficient (female minus male) should therefore be interpreted as the residual group association after adjustment for age, total volume, and these three coarse anatomical descriptors; it is not a claim of complete anatomical conditioning. We report 95% CIs, standardized coefficients, partial R^2 , and BH-FDR corrected p values.

Pooled robust models across load cases (H1/H2/H4)

Because each subject contributed all five loads, we used cluster-robust OLS (clustered by subject ID) for each primary outcome:

$$Y \sim \text{load case} + \text{sex} + \text{age}_z + \log(\text{volume})_z + \text{AP}_z + \text{Biischiadic}_z + \text{Subpubic}_z \quad (3)$$

$$+ \text{load case} \times \log(\text{volume})_z + \text{sex} \times \log(\text{volume})_z. \quad (4)$$

This model quantified load contrasts against SP2leg, adjusted sex effects, and load-dependent allometric modulation while accounting for repeated measures within subjects.

Variance-channel decomposition (H3)

For each load and SIJ metric, variance was computed in three matched simulation repositories: full, shape-only, and material-only. In the shape-only repository, the subject mapping ϕ was applied while the material mapping was held at the reference assignment; in the material-only repository, subject-specific material mapping was applied on the reference geometry (i.e., $\phi = 0$). We reported shape/full and material/full variance ratios (%) with bootstrap 95% CIs (Efron and Tibshirani, 1994). Because these are ratios of sample variances from separate simulation variants rather than partitions of a single variance term, values slightly above 100% can occur and should be interpreted as near-equality rather than literal “more than full” variance.

Size-scaling models (H4, Variant B)

For each load and each primary outcome, we fit robust log–log models:

$$\log(Y) \sim \log(\text{total volume}) + \text{sex} + \text{age}_z + \text{AP}_z + \text{Biischadic}_z + \text{Subpubic}_z + \log(\text{total volume}) \times \text{sex}. \quad (5)$$

Male and female scaling exponents were extracted from main and interaction terms. A supplementary sensitivity analysis replaced $\log(\text{total volume})$ with $\log(\text{scale})$.

2.6 Multiplicity and software

False-discovery control used Benjamini–Hochberg correction within each analysis family (Benjamini and Hochberg, 1995). We report corrected p values, 95% CIs, standardized betas, and partial R^2 . Analyses were run in Python (pandas, statsmodels, scipy, matplotlib, zarr).

3 Results

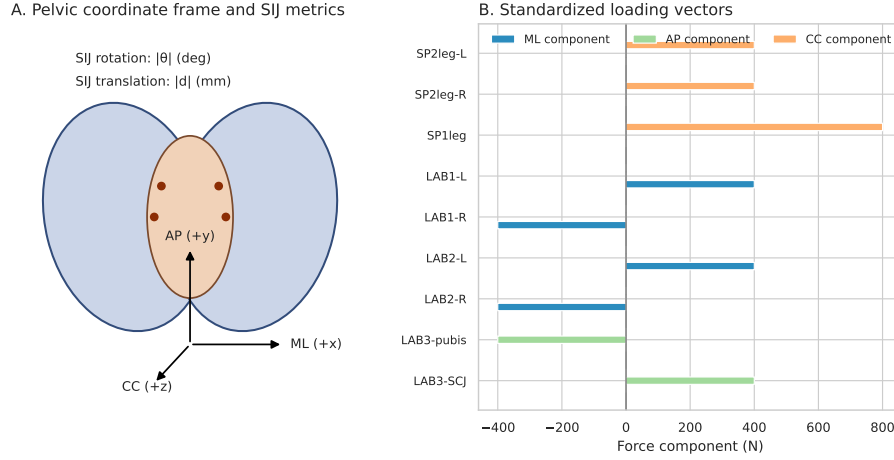


Figure 1: Anatomy schematic, pelvic coordinate definition, SIJ metric definitions, and standardized load vectors.

3.1 Cohort completeness and baseline morphology

The analyzed cohort comprised 278 subjects (150 female, 128 male), with age 53.9 ± 19.3 years (range 16–91). No missing values were present for demographic variables, pelvic dimensions, SIJ primary outcomes, or global size-proxy metrics (Table 1). Female pelves showed larger mean inlet AP diameter (121.0 vs 112.3 mm), biischiadic width (109.9 vs 97.7 mm), and subpubic angle (85.8 vs 69.1°) than male pelves, but smaller global scale factor and total pelvic volume.

3.2 Load-case dependence of SIJ micromotion (H1)

Median SIJ micromotion varied strongly by load (Table 3; Fig. 2). SP1leg produced the largest response (rotation 2.516 [1.981 , 3.137] deg; translation 1.755 [1.458 , 2.053] mm), exceeding SP2leg (1.692 [1.211 , 2.451] deg; 1.299 [0.971 , 1.625] mm). Relative to SP2leg, LAB1 and LAB2 reduced overall translation magnitude (0.665 and 0.764 mm medians), while LAB3 showed higher rotation than LAB1/LAB2 (1.769 deg) with low translation (0.790 mm median).

Cluster-robust pooled models confirmed these contrasts. For rotation magnitude, coefficients versus SP2leg were $+0.723$ deg for SP1leg, -0.326 deg for LAB1, -0.261 deg for LAB2, and $+0.060$ deg for LAB3 (all BH-FDR < 0.001). For translation magnitude, coefficients were $+0.499$ mm (SP1leg), -0.598 mm (LAB1), -0.498 mm (LAB2), and -0.494 mm (LAB3) (all BH-FDR $< 10^{-20}$).

Table 3: SIJ micromotion summary by load case. Values are median [IQR].

Load case	N	Rotation magnitude (deg)	Translation magnitude (mm)	AP translation component (mm)	ML translation component (mm)	CC translation component (mm)
SP2leg	278	1.692 [1.211, 2.451]	1.299 [0.971, 1.625]	0.272 [0.158, 0.433]	0.318 [0.160, 0.574]	1.131 [0.855, 1.459]
SP1leg	278	2.516 [1.981, 3.137]	1.755 [1.458, 2.053]	0.431 [0.299, 0.572]	0.341 [0.200, 0.550]	1.527 [1.262, 1.775]
LAB1	278	1.393 [0.908, 2.060]	0.665 [0.450, 0.923]	0.268 [0.169, 0.424]	0.410 [0.228, 0.678]	0.263 [0.163, 0.410]
LAB2	278	1.433 [1.011, 2.069]	0.764 [0.536, 1.048]	0.413 [0.216, 0.628]	0.476 [0.284, 0.738]	0.236 [0.138, 0.377]
LAB3	278	1.769 [1.169, 2.591]	0.790 [0.568, 1.052]	0.262 [0.143, 0.473]	0.274 [0.138, 0.511]	0.534 [0.325, 0.749]

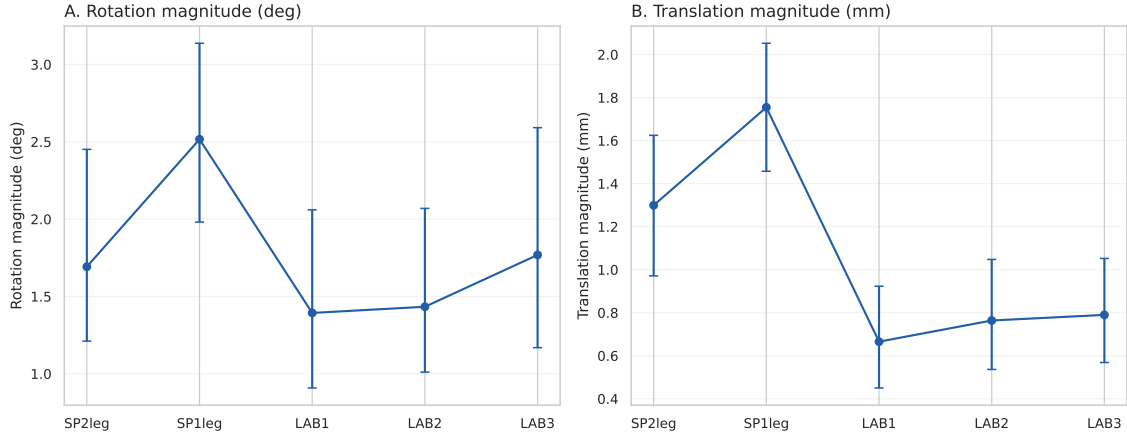


Figure 2: Primary SIJ outcomes across load cases (medians with IQR error bars).

3.3 Directional re-routing rather than uniform scaling (H1)

Directional translation differences versus SP2leg showed load-specific anisotropy (Fig. 3; Supplementary Table S1). SP1leg increased AP (+0.072 mm, 95% CI 0.045 to 0.096, BH-FDR 1.31×10^{-22}) and CC (+0.394 mm, 0.347 to 0.434, BH-FDR 4.73×10^{-43}) components, while ML change was small and not significant after correction.

LAB loads exhibited clear re-routing. LAB1 increased ML (+0.086 mm, 0.077 to 0.095) but strongly reduced CC (-0.850 mm, -0.881 to -0.811), with no AP change after correction. LAB2 increased AP (+0.163 mm, 0.102 to 0.213) and ML (+0.145 mm, 0.132 to 0.160), with strong CC reduction (-0.903 mm, -0.931 to -0.853). LAB3 showed reduced ML (-0.041 mm, -0.052 to -0.033) and reduced CC (-0.586 mm, -0.609 to -0.544), while AP remained unchanged.

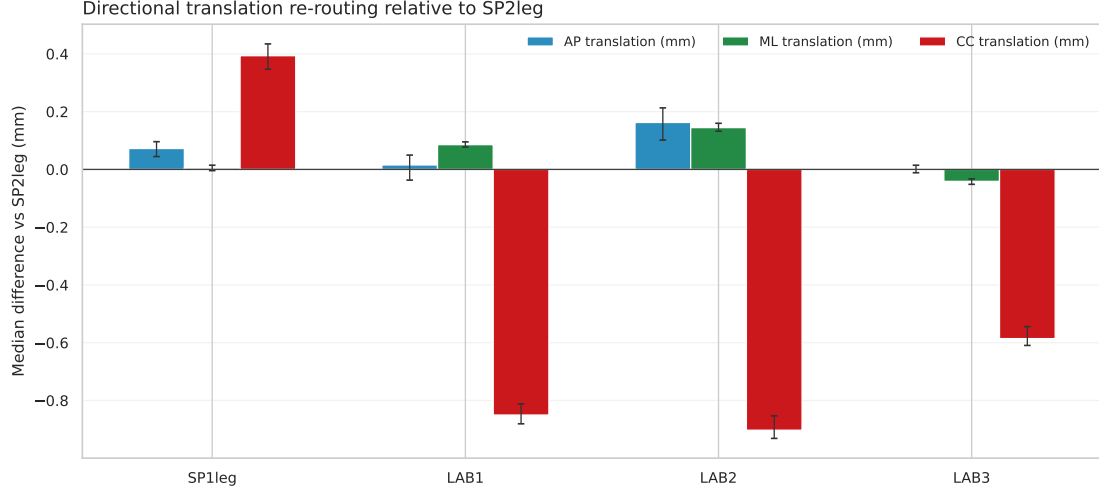


Figure 3: Directional translation re-routing relative to SP2leg. Bars show median within-subject differences with 95% bootstrap CIs.

3.4 Sex effects under parturition-motivated loads (H2)

Adjusted LAB-phase robust models showed higher female micromotion after coarse anatomical adjustment for most primary outcomes (Fig. 4; Supplementary Table S3). For translation magnitude, female effects were +0.229 mm in LAB1 (95% CI 0.078 to 0.379; BH-FDR 0.007), +0.199 mm in LAB2 (0.039 to 0.359; BH-FDR 0.022), and +0.233 mm in LAB3 (0.078 to 0.388; BH-FDR 0.007). For rotation magnitude, evidence was narrower: the female effect was strongest in LAB3 (+0.627 deg, 0.206 to 1.048; BH-FDR 0.007), whereas LAB1 was borderline after correction (BH-FDR 0.054) and LAB2 was not significant.

In pooled cluster-robust models across all loads, female sex remained associated with higher rotation (+0.450 deg, BH-FDR 0.020) and translation (+0.254 mm, BH-FDR 3.41×10^{-4}), indicating that the group difference was not confined to a single LAB phase.

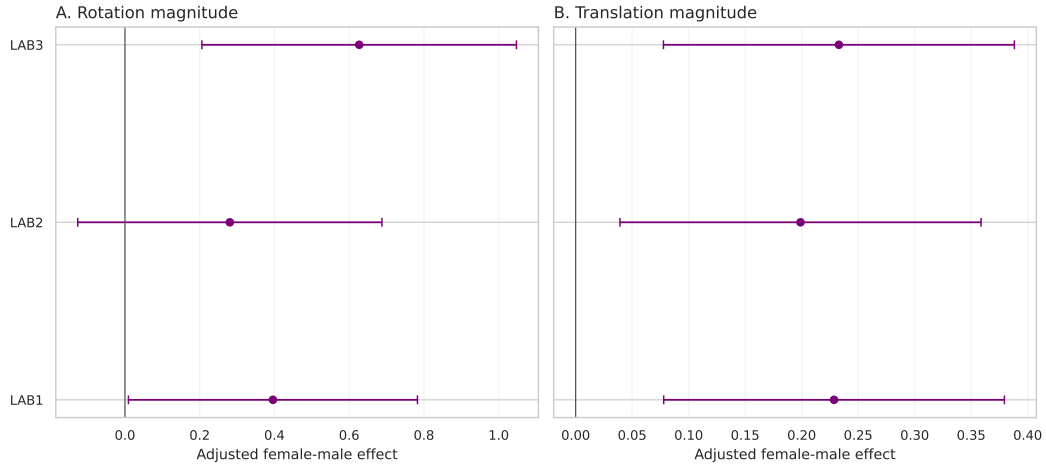


Figure 4: Adjusted female-minus-male effects for primary outcomes in LAB1–LAB3 (95% CI).

3.5 Variance channels: shape-dominant variability (H3)

Variance-channel comparison indicated geometry-dominant *between-subject* variability. Shape-only simulations reproduced full-model variance to within a narrow margin (98.7–102.1% across load/metric combinations; median 100.46%), while material-only variability was orders of magnitude smaller (0.001–0.317%; median 0.033%) (Fig. 5; Supplementary Table S2). Ratios slightly above 100% arise because these are sample-variance ratios computed on separate simulation variants; values near 100% therefore indicate near-complete retention rather than a paradoxical “excess” of variance.

For LAB phases specifically, shape/full ratios were close to 100% for all primary and directional metrics, while material/full remained below 0.28% in all LAB combinations. This supports dominant geometry-driven inter-subject SIJ variability under standardized loading.

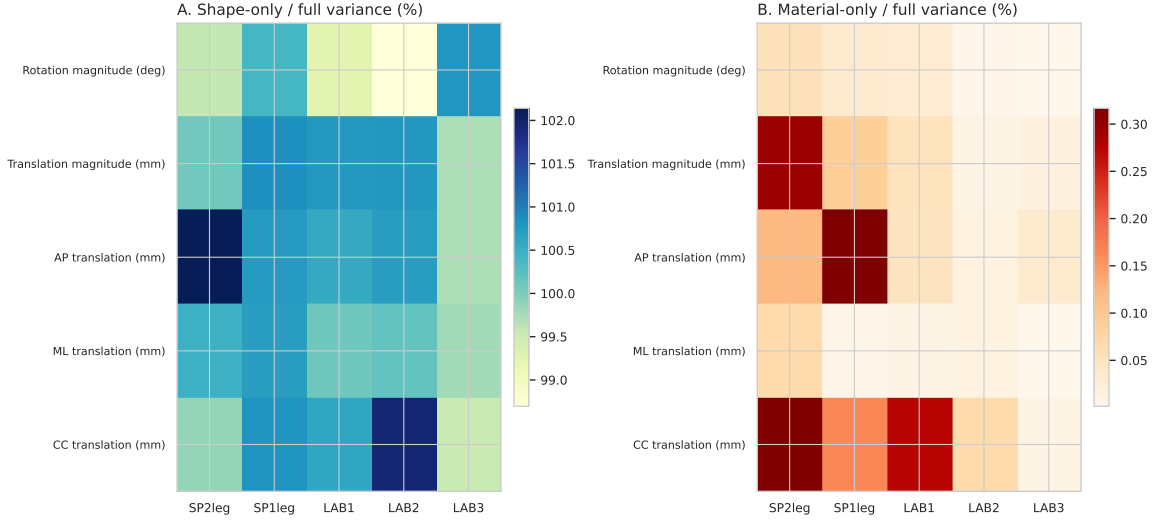


Figure 5: Variance-channel ratios across load cases and SIJ metrics.

3.6 Scaling of SIJ micromotion with global pelvic size (H4)

Load-wise log-log allometry models are summarized in Table 4. Evidence for negative scaling was clearest under SP1leg: larger pelvic volume was associated with lower micromotion in both sexes, with rotation exponents of -0.482 (male, 95% CI -0.870 to -0.094) and -0.428 (female, -0.762 to -0.093), and translation exponents of -0.569 (male, -0.869 to -0.269) and -0.432 (female, -0.669 to -0.194). In contrast, SP2leg exponents had confidence intervals crossing zero, and several LAB-phase exponents were likewise imprecise.

Sex-by-size interaction terms for allometry slopes were not significant after BH-FDR correction in any load/outcome pair (all corrected $p \geq 0.958$), indicating no strong evidence for sex-divergent scaling exponents in this dataset.

In pooled cluster-robust models, translation magnitude showed an overall negative size association (standardized log-total-volume coefficient -0.092 per 1 SD, 95% CI -0.143 to -0.041; BH-FDR 6.17×10^{-4}), with significant load-dependent modulation through load $\times$ size interactions. Rotation showed a weaker pooled size signal, consistent with the broader uncertainty in the load-wise rotation exponents.

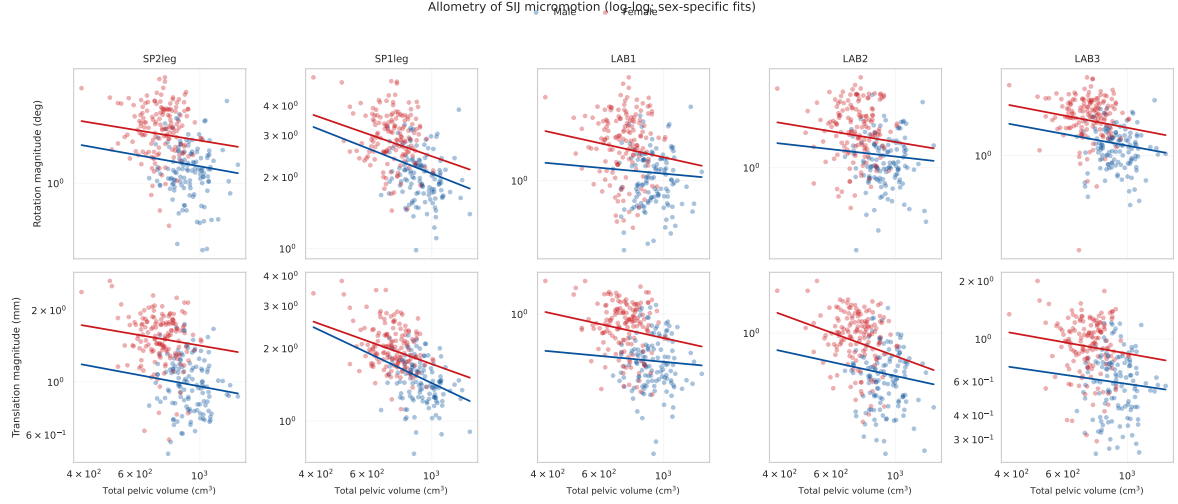


Figure 6: Log-log allometry of primary SIJ outcomes versus total pelvic volume, stratified by sex and load case.

Table 4: Primary robust regression and size-scaling results. Panel A: pooled cluster-robust primary models. Panel B: load-wise scaling exponents.

Panel A

outcome	term	beta	ci95_low	ci95_high	beta_std	partial_r2	p_fdr_bh
rot_mag_deg	C(load_case, Treatment(reference='SP2leg'))[T.SP1leg]	0.723	0.672	0.773	0.781	0.362	0.000
rot_mag_deg	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase1]	-0.326	-0.359	-0.292	-0.352	0.209	0.000
rot_mag_deg	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase2]	-0.261	-0.323	-0.198	-0.282	0.046	0.000
rot_mag_deg	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase3]	0.060	0.033	0.087	0.065	0.013	0.000
rot_mag_deg	sex_F	0.450	0.103	0.797	0.487	0.005	0.020
rot_mag_deg	age_z	0.044	-0.053	0.141	0.048	0.001	0.398
rot_mag_deg	log_total_volume_z	-0.126	-0.285	0.034	-0.136	0.002	0.182
rot_mag_deg	C(load_case, Treatment(reference='SP2leg'))[T.SP1leg]:log_total_volume_z	-0.032	-0.079	0.015	-0.035	0.001	0.223
rot_mag_deg	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase1]:log_total_volume_z	0.096	0.062	0.130	0.104	0.022	0.000
rot_mag_deg	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase2]:log_total_volume_z	0.105	0.033	0.177	0.114	0.006	0.009
rot_mag_deg	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase3]:log_total_volume_z	-0.065	-0.093	-0.038	-0.071	0.016	0.000
rot_mag_deg	AP_z	0.130	0.028	0.232	0.141	0.005	0.020
rot_mag_deg	BiischadicWidth_z	0.082	-0.035	0.200	0.089	0.001	0.223
rot_mag_deg	SubpubicAngle_z	0.079	-0.075	0.232	0.085	0.001	0.366
rot_mag_deg	sex_F:log_total_volume_z	-0.079	-0.280	0.123	-0.085	0.000	0.443
trans_mag_mm	C(load_case, Treatment(reference='SP2leg'))[T.SP1leg]	0.499	0.465	0.533	0.884	0.375	0.000
trans_mag_mm	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase1]	-0.598	-0.627	-0.569	-1.060	0.542	0.000
trans_mag_mm	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase2]	-0.498	-0.533	-0.462	-0.882	0.353	0.000
trans_mag_mm	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase3]	-0.494	-0.513	-0.475	-0.875	0.659	0.000
trans_mag_mm	sex_F	0.254	0.121	0.387	0.450	0.010	0.000
trans_mag_mm	age_z	0.028	-0.008	0.063	0.049	0.002	0.134
trans_mag_mm	log_total_volume_z	-0.092	-0.143	-0.041	-0.163	0.009	0.001
trans_mag_mm	C(load_case, Treatment(reference='SP2leg'))[T.SP1leg]:log_total_volume_z	-0.073	-0.106	-0.040	-0.129	0.013	0.000
trans_mag_mm	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase1]:log_total_volume_z	0.073	0.043	0.103	0.130	0.016	0.000
trans_mag_mm	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase2]:log_total_volume_z	0.048	0.008	0.088	0.085	0.004	0.025
trans_mag_mm	C(load_case, Treatment(reference='SP2leg'))[T.LAB_phase3]:log_total_volume_z	0.085	0.068	0.102	0.150	0.063	0.000
trans_mag_mm	AP_z	-0.062	-0.095	-0.028	-0.109	0.009	0.000
trans_mag_mm	BiischadicWidth_z	0.043	-0.002	0.089	0.076	0.002	0.079
trans_mag_mm	SubpubicAngle_z	0.051	-0.012	0.114	0.091	0.002	0.129
trans_mag_mm	sex_F:log_total_volume_z	-0.039	-0.111	0.034	-0.069	0.001	0.293

Panel B

load_case	outcome	male_exponent	male_ci95_low	male_ci95_high	female_exponent	female_ci95_low	female_ci95_high	sex_slope_interaction_beta	interaction_p_fdr_bh	adj_r2
SP2leg	rot_mag_deg	-0.334	-0.931	0.263	-0.306	-0.762	0.151	0.028	0.958	0.383
SP2leg	trans_mag_mm	-0.227	-0.562	0.107	-0.211	-0.505	0.084	0.017	0.958	0.468
SP1leg	rot_mag_deg	-0.482	-0.870	-0.094	-0.428	-0.762	-0.093	0.055	0.958	0.382
SP1leg	trans_mag_mm	-0.569	-0.869	-0.269	-0.432	-0.669	-0.194	0.137	0.958	0.500
LAB_phase1	rot_mag_deg	-0.172	-0.846	0.502	-0.416	-0.959	0.127	-0.244	0.958	0.267
LAB_phase1	trans_mag_mm	-0.177	-0.665	0.311	-0.419	-0.914	0.077	-0.242	0.958	0.354
LAB_phase2	rot_mag_deg	-0.228	-1.020	0.565	-0.334	-0.891	0.223	-0.106	0.958	0.269
LAB_phase2	trans_mag_mm	-0.367	-0.904	0.171	-0.614	-1.061	-0.166	-0.247	0.958	0.375
LAB_phase3	rot_mag_deg	-0.498	-1.169	0.174	-0.518	-1.048	0.011	-0.021	0.958	0.367
LAB_phase3	trans_mag_mm	-0.220	-0.720	0.280	-0.271	-0.708	0.166	-0.051	0.958	0.310

3.7 Summary relative to hypotheses

Taken together, the results provide the following hypothesis-level interpretation. **H1** was supported: SIJ micromotion differed across load cases, and parturition-motivated loads produced directional re-routing rather than uniform scaling (Table 3; Figs. 2–3). **H2** was supported in a qualified sense: after adjustment for age, total volume, and three interpretable pelvic dimensions, female-associated increases were most consistent for LAB-phase translation magnitude and were strongest for LAB3 rotation (Fig. 4). **H3** was strongly supported: between-subject variability was largely reproduced by shape-only simulations, while material-only fractions were small (Fig. 5). **H4** was partially supported: size scaling was load dependent, with the clearest negative exponents under one-leg standing and in pooled translation models, but sex-by-size slope differences were not supported after multiplicity control (Fig. 6; Table 4).

4 Discussion

This study used a large, precomputed FE repository to characterize SIJ micromotion under standardized standing and parturition-motivated loading with explicit anatomy and global size-proxy covariates derived from an implicit geometry mapping. The analysis was structured around four pre-specified hypotheses (H1–H4) concerning load dependence and directional re-routing, sex effects under parturition-motivated loading, variance channels, and size scaling. Two general points frame the interpretation. First, the model addresses standardized static loading, so the present results speak most directly to comparative load routing across subjects and load cases rather than to the full time-varying physiology of gait or labor. Second, the strongest geometry result concerns *between-subject dispersion*: geometry dominates how much subjects differ from one another under a fixed loading protocol, whereas the largest shifts in scalar micromotion across the paper are still driven by load case itself.

Interpretation by hypothesis

H1 (load dependence and re-routing). SIJ response was strongly load dependent and direction specific. One-leg standing generated the largest overall SIJ micromotion, consistent with unilateral support demands and with the broader literature showing that routine activities can impose markedly different hip and pelvic loading states (Bergmann et al., 2001; Kutzner et al., 2016). In contrast, the parturition-motivated load proxies produced lower global transla-

tion magnitude than standing but clear directional re-routing: LAB1 shifted motion toward the ML component, and LAB2 toward both ML and AP components, while both markedly reduced CC translation; LAB3 preserved CC reduction with a distinct ML pattern. These observations support an anisotropic interpretation of SIJ mechanics in which load direction changes the partitioning of motion across AP/ML/CC channels rather than scaling micromotion uniformly (Vleeming et al., 2012; Goode et al., 2008; Meijer et al., 2014; Keizer et al., 2019; No"e et al., 2020).

The absolute magnitudes predicted here also remain compatible with the broader picture of the SIJ as a low-motion but load-sensitive articulation. Reviews of SIJ kinematics emphasize both small overall motion and substantial dependence on task and measurement approach, while functional-anatomy work highlights that load distribution and joint position influence the direction and amount of allowed motion (Goode et al., 2008; Vleeming et al., 2012; Meijer et al., 2014). We therefore interpret the present FE outputs less as direct surrogates for any single *in vivo* experiment and more as a mechanically standardized map of how different loading patterns redistribute motion within the pelvic ring.

H2 (sex effects conditional on anatomy). Under parturition-motivated loading, female sex was associated with higher adjusted SIJ micromotion, most consistently for translation magnitude across LAB phases and for rotation magnitude in the LAB3 proxy. Because these effects persisted after adjustment for age, total pelvic volume, and three interpretable pelvic dimensions, they are unlikely to be explained solely by overall body size. Instead, they are consistent with sex-dimorphic pelvic architecture shaping compliance pathways and load routing through the pelvic ring (Ruff, 2017; Weiss, 2020). However, this remains a *coarse* anatomical adjustment, not an exhaustive conditioning on shape: the models do not explicitly encode joint surface curvature, sacral auricular orientation, or ligament attachment geometry. The sex coefficient should therefore be interpreted as a residual group association after limited anatomical conditioning, not as evidence for a single latent sex-specific mechanism.

H3 (geometry-dominant variability). Between-subject variability in SIJ outcomes was geometry dominant. Shape-only outputs reproduced essentially all full-model variance, whereas material-only variance fractions were very small. Values slightly above 100% for the shape/full ratio are not paradoxical: they reflect ratios of sample variances computed on separate simulation variants, not a literal super-additive variance source. Under the present force-standardized

loading protocol, these findings indicate that subject-specific morphology and geometry carry most of the inter-subject dispersion signal, while material-only perturbation contributes comparatively little to spread in the tested outcomes.

This result should be read carefully. It does *not* mean that material properties are irrelevant to SIJ mechanics in general; rather, within this repository and for these outputs, geometry explains far more of why subjects differ from one another than the isolated material-only perturbation does. That interpretation is compatible with prior FE work showing sensitivity of pelvic response to ligament restraint, cartilage properties, and loading configuration (Anderson et al., 2020; Yang et al., 2020; Song et al., 2021; Calixte et al., 2022). Our analysis adds that, once loading is standardized across a large cohort, the dominant source of inter-subject spread resides in mapped anatomy.

H4 (size scaling). Size scaling was load dependent, but the evidence was more specific than a blanket “larger pelvis, less motion” statement. The clearest negative exponents occurred under one-leg standing, and pooled translation models also showed a negative volume association. By contrast, bilateral-standing exponents were imprecise and several LAB-specific exponents crossed zero. Thus, the data support a conditional interpretation: negative scaling is most evident when the loading pattern creates larger unilateral demand or when translation is pooled across load cases, not uniformly across every condition.

This pattern is mechanically plausible in a force-standardized setting, where larger load-bearing geometry may distribute identical resultant loads through broader structural pathways. At the same time, the weak sex-by-size interactions indicate that the dataset does not support strongly different scaling laws for women and men once volume and the coarse morphometric covariates are included. For the current manuscript, the safest conclusion is therefore that allometry modulates SIJ micromotion, but its strength and direction are load dependent.

Anatomical and functional implications

The combined findings support an anatomy-first interpretation of SIJ mechanics, but with an important distinction between *routing* and *magnitude*. Load direction governs how motion is partitioned across AP/ML/CC channels, whereas mapped pelvic geometry governs most of the between-subject spread under a fixed load definition. From an obstetric-mechanical perspective, this suggests that evaluating only scalar mobility may miss clinically relevant directional

behavior, particularly under parturition-motivated loading states.

The observed sex differences in adjusted micromotion, together with established sex-related contrasts in pelvic morphology (Ruff, 2017; Betti and Manica, 2020; Weber et al., 2019; Whitcome et al., 2017; Weiss, 2020), support the view that group-level differences emerge from load-case-specific routing through a sexually dimorphic pelvic ring rather than from a uniform “more mobile” versus “less mobile” SIJ phenotype. In other words, the data favor a routing-based interpretation of sex dimorphism: women and men differ not only in how much motion is produced under some loads, but also in how that motion is distributed across anatomical directions when the external load vector is changed.

Limitations

The analysis remains constrained by the assumptions of the underlying repository. The mechanical model is linear, small-strain, and static; it does not include time-varying contractions, contact-rich fetal-head interactions, or hormonally mediated tissue adaptation (Chen and Grimm, 2021; Roque de Almeida et al., 2024; House et al., 2022; Mora et al., 2022). The LAB load cases are therefore proxies, not full labor reconstructions. Material mapping follows a fixed density–modulus law and fixed cartilage/symphysis parameters. In addition, although age was adjusted statistically, the model does not explicitly encode age-dependent tissue remodeling trajectories (Farr and Khosla, 2015; Zebaze et al., 2010).

Two further limitations are specific to the present statistical framing. First, the anatomical adjustment set is intentionally parsimonious, so residual sex effects cannot be interpreted as fully anatomy-independent. Second, the variance-channel analysis compares separate simulation repositories; it cleanly addresses relative spread under standardized protocols, but it is not a formal variance decomposition of a single stochastic generative model. These boundaries should temper causal interpretation.

Outlook

Future extensions should combine this population-scale geometry framework with nonlinear contact models and subject-specific soft-tissue updates across pregnancy. A second priority is external validation against *in vivo* SIJ kinematic datasets and imaging-derived pelvic-ring deformation measurements in matched loading scenarios. More generally, the present workflow suggests a useful division of labor for future studies: use large standardized FE cohorts to identify

robust routing signatures and variance channels, then test those signatures experimentally in smaller, higher-fidelity cohorts.

5 Conclusion

In a 278-subject pelvic FE cohort represented on a common template via an implicit geometry mapping, SIJ micromotion under standardized loads showed three robust features: strong load dependence, clear directional re-routing under parturition-motivated proxies, and geometry-dominant between-subject variability. Female-associated increases in LAB-phase micromotion persisted after adjustment for age, total pelvic volume, and three interpretable pelvic dimensions, but these residual associations should be read as surviving a coarse rather than exhaustive anatomical adjustment. Size scaling was load dependent, with the clearest negative associations under one-leg standing and in pooled translation models, and without robust evidence for sex-divergent scaling slopes. Together, these results support an anatomy-centered view of SIJ mechanics in which load direction governs motion partitioning and pelvic geometry governs most of the inter-subject spread under a standardized loading protocol.

Acknowledgements

We thank the BoneDat project for curating and providing the standardized imaging-derived dataset used in this study.

Funding

This work was supported by the Czech Science Foundation (GAČR; Grant No. 24-10862S).

Author contributions

CRedit authorship contribution statement. Petr Henyš: Conceptualization; Methodology; Software; Formal analysis; Investigation; Data curation; Visualization; Funding acquisition; Writing – original draft; Writing – review & editing. Niels Hammer: Supervision; Writing – review & editing. All authors reviewed and approved the final manuscript.

Conflict of interest statement

The authors declare no conflicts of interest.

Supporting information

Supporting Information includes Supplementary Methods (implicit mapping, load pullback, SIJ kinematics, and size-scaling model details) and Supplementary Tables S1–S4.

Data Availability Statement

The derived data supporting the findings of this study (summary tables and figures) are provided with the manuscript. Subject-level derived datasets and analysis outputs are available in a blinded repository for peer review and will be made publicly available upon publication.

Code Availability Statement

All custom code used to post-process the finite-element outputs and perform the statistical analyses is available in the same blinded repository for peer review and will be made publicly available upon publication.

References

- Anderson, A. et al. (2020). Role of sacroiliac ligaments in linear stiffness: a finite element study. *Journal of Orthopaedic Research*, 38(9):2027–2036.
- Benjamini, Y. and Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300.
- Bergmann, G., Deuretzbacher, F., Heller, M., Graichen, V., Rohlmann, A., Strauss, J., and Duda, G. N. (2001). Hip contact forces and gait patterns from routine activities. *Journal of Biomechanics*, 34(7):859–871.
- Betti, L. and Manica, A. (2020). Human variation in the shape of the bony pelvis and climatic adaptation. *Nature Ecology & Evolution*, 4:153–160.

456 Bonet, J. and Wood, R. D. (1997). *Nonlinear Continuum Mechanics for Finite Element Analysis*.
457 Cambridge University Press, 1st edition.

458 Calixte, R. et al. (2022). Cartilage properties modulate sacroiliac joint linear response: finite
459 element evidence. *Journal of Biomechanics*, 135:111022.

460 Carter, D. R. and Hayes, W. C. (1977). The compressive behavior of bone as a two-phase porous
461 structure. *The Journal of Bone and Joint Surgery. American Volume*, 59(7):954–962.

462 Chen, S. and Grimm, M. J. (2021). Childbirth computational models: Characteristics and
463 applications. *Journal of Biomechanical Engineering*, 143(5):050801.

464 Chun, R. and Kirkpatrick, A. W. (2012). Intra-abdominal pressure, intra-abdominal hyperten-
465 sion, and pregnancy: a review. *Annals of Intensive Care*, 2(S1):S5.

466 Efron, B. and Tibshirani, R. J. (1994). *An Introduction to the Bootstrap*. Chapman & Hall/CRC.

467 Farr, J. N. and Khosla, S. (2015). Skeletal changes through the lifespan—from growth to
468 senescence. *Nature Reviews Endocrinology*, 11(9):513–521.

469 Frost, H. M. (1987). Bone ‘mass’ and the ‘mechanostat’: A proposal. *The Anatomical Record*,
470 219(1):1–9.

471 Fuchs, F., Bruyère, M., Senat, M.-V., Purenne, E., Benhamou, D., and Fernandez, H. (2013).
472 Are standard intra-abdominal pressure values different during pregnancy? *PLOS ONE*,
473 8(10):e77324.

474 Goode, A., Hegedus, E. J., and Sizer, P. (2008). Three-dimensional movements of the sacroiliac
475 joint: A systematic review of the literature and assessment of clinical utility. *Journal of*
476 *Manual & Manipulative Therapy*, 16(1):25–38.

477 Gruss, L. T. and Schmitt, D. (2015). The evolution of the human pelvis: changing adaptations
478 to bipedalism? *The Anatomical Record*, 298(7):1235–1245.

479 Henyš, P. and Kuchař, M. (2025). Bonedat, a database of standardized bone morphology for in
480 silico analyses. *Scientific Data*, 12(1).

481 House, M., Garfield, J., Elovitz, M. A., et al. (2022). Assessment of uterine contractions in labor
482 and delivery: current challenges and future directions. *American Journal of Obstetrics and*
483 *Gynecology*, 227(5):B27–B41.

484 Huiskes, R., Weinans, H., Grootenboer, H. J., Dalstra, M., Fudala, B., and Slooff, T. J. J. H.
 485 (1987). Adaptive bone-remodeling theory applied to prosthetic-design analysis. *Journal of*
 486 *Biomechanics*, 20(11–12):1135–1150.

487 Keizer, R. et al. (2019). In-vivo sacroiliac joint motion with biplanar radiography. *Journal of*
 488 *Biomechanics*, 92:7–13.

489 Keller, T. S. (1994). Predicting the compressive mechanical behavior of bone. *Journal of*
 490 *Biomechanics*, 27(9):1159–1168.

491 Kutzner, I., Trepczynski, A., Heller, M. O., and Bergmann, G. (2016). Standardized loads acting
 492 in hip implants. *PLOS ONE*, 11(5):e0155612.

493 MacKinnon, J. G. and White, H. (1985). Some heteroskedasticity-consistent covariance matrix
 494 estimators with improved finite sample properties. *Journal of Econometrics*, 29(3):305–325.

495 Marsden, J. E. and Hughes, T. J. R. (1983). *Mathematical Foundations of Elasticity*. Prentice
 496 Hall.

497 Meijer, O. et al. (2014). Sacroiliac joint motion: systematic review and meta-analysis. *The*
 498 *Spine Journal*, 14(9):1926–1941.

499 Mora, I., Lucia, A., Martínez-González, J., et al. (2022). Biomechanics of pregnancy and vaginal
 500 delivery. *Biomechanics and Modeling in Mechanobiology*, 21(5):1647–1670. Review reporting
 501 vaginal wall–fetal head pressures up to ~ 32 kPa during pushing.

502 Morgan, E. F., Bayraktar, H. H., and Keaveny, T. M. (2003). Trabecular bone modulus-density
 503 relationships depend on anatomic site. *Journal of Biomechanics*, 36(7):897–904.

504 No"e, C. et al. (2020). In vivo sacroiliac joint motion using biplanar imaging. *Clinical Biome-*
 505 *chanics*, 80:105115.

506 Roque de Almeida, P., Alves Campice de Oliveira, I., de Oliveira Campos, J., Martins Rocha,
 507 B., and de Souza Bastos, F. (2024). Modeling of the biomechanical behavior and growth of
 508 the human uterus during pregnancy. *Journal of Biomechanics*, 174:112268.

509 Ruff, C. (2017). Morphological variation of the human pelvis and its functional implications.
 510 *American Journal of Physical Anthropology*, 162(S63):32–71.

511 Snijders, C. J., Vleeming, A., and Stoeckart, R. (1993). Transfer of lumbosacral load to iliac
512 bones and legs. part 1: Biomechanics of self-bracing of the sacroiliac joints. *Clinical Biome-*
513 *chanics*, 8(6):285–294.

514 Song, Y. et al. (2021). Pelvic ring biomechanics under physiological loads: finite element study.
515 *Medical Engineering & Physics*, 93:1–10.

516 Vleeming, A., Schuenke, M. D., Masi, A. T., Carreiro, J. E., Danneels, L., and Willard, F. H.
517 (2012). The sacroiliac joint: an overview of its anatomy, function and potential clinical
518 implications. *Journal of Anatomy*, 221(6):537–567.

519 Washburn, S. L. (1960). Tools and human evolution. *Scientific American*, 203(3):62–75.

520 Weber, G. W. et al. (2019). Virtual anthropology: Morphometric variation of the human pelvis
521 across populations. *Scientific Reports*, 9:19112.

522 Weiss, E. (2020). Sex and ancestry effects on pelvic shape. *Journal of Anatomy*, 236:109–126.

523 Whitcome, K. K. et al. (2017). Evolutionary biomechanics of the human pelvis. *Philosophical*
524 *Transactions of the Royal Society B*, 372(1720):20160021.

525 White, H. (1980). A heteroskedasticity-consistent covariance matrix estimator and a direct test
526 for heteroskedasticity. *Econometrica*, 48(4):817–838.

527 Wilcoxon, F. (1945). Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–
528 83.

529 Yang, X. et al. (2020). Effects of ligaments on pelvic stability: 3d finite element analysis. *Journal*
530 *of Biomechanical Engineering*, 142(8).

531 Zebaze, R. M., Ghasem-Zadeh, A., Bohte, A., Iuliano-Burns, S., Mirams, M., Price, R. I.,
532 Mackie, E. J., and Seeman, E. (2010). Intracortical remodelling and porosity in the distal ra-
533 dius and post-mortem femurs of women: a cross-sectional study. *The Lancet*, 375(9727):1729–
534 1736.